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Novak et al.

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[45] **Date of Patent: May 5, 1987**

[54] **RANDOM/SERIAL ACCESS MODE SELECTION CIRCUIT FOR A VIDEO MEMORY SYSTEM**

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[73] Assignee: **Texas Instruments Incorporated,** Dallas, Tex.

[21] Appl. No.: **567,111**

[22] Filed: **Dec. 30, 1983**

[51] **Int. Cl.**[4] **G06F 13/00;** G11C 8/00

[52] **U.S. Cl.** **364/900;** 365/233; 365/189; 340/750

[58] **Field of Search** ... 364/200 MS File, 900 MS File, 364/521; 365/219, 220, 221, 239, 230, 233, 189; 340/750, 798, 799, 800, 801

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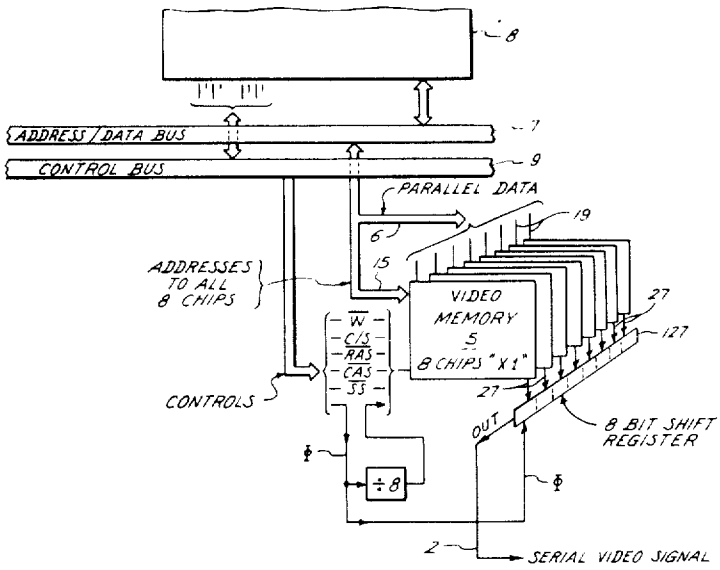
Assistant Examiner—Florin Munteanu
Attorney, Agent, or Firm—John G. Graham; Rodney M. Anderson

[57] **ABSTRACT**

In a video computer system, an improved memory circuit is provided which is effective for delivering stored data only at appropriate instances, and which is also simpler and more reliable in design. In particular, the system preferably includes a bit-mapped RAM circuit which assumes a serial mode in response to both a row address signal and a suitable data output control signal, and which assumes a parallel or "random" mode when only the row address is received. Stored data is transferred to a parallel output terminal in the RAM circuit, or to a serial output terminal therein, depending upon the sequence of these signals as well as the column address and read signals, whereby the data output control signal is used for two separate and different purposes within the system.

7 Claims, 13 Drawing Figures



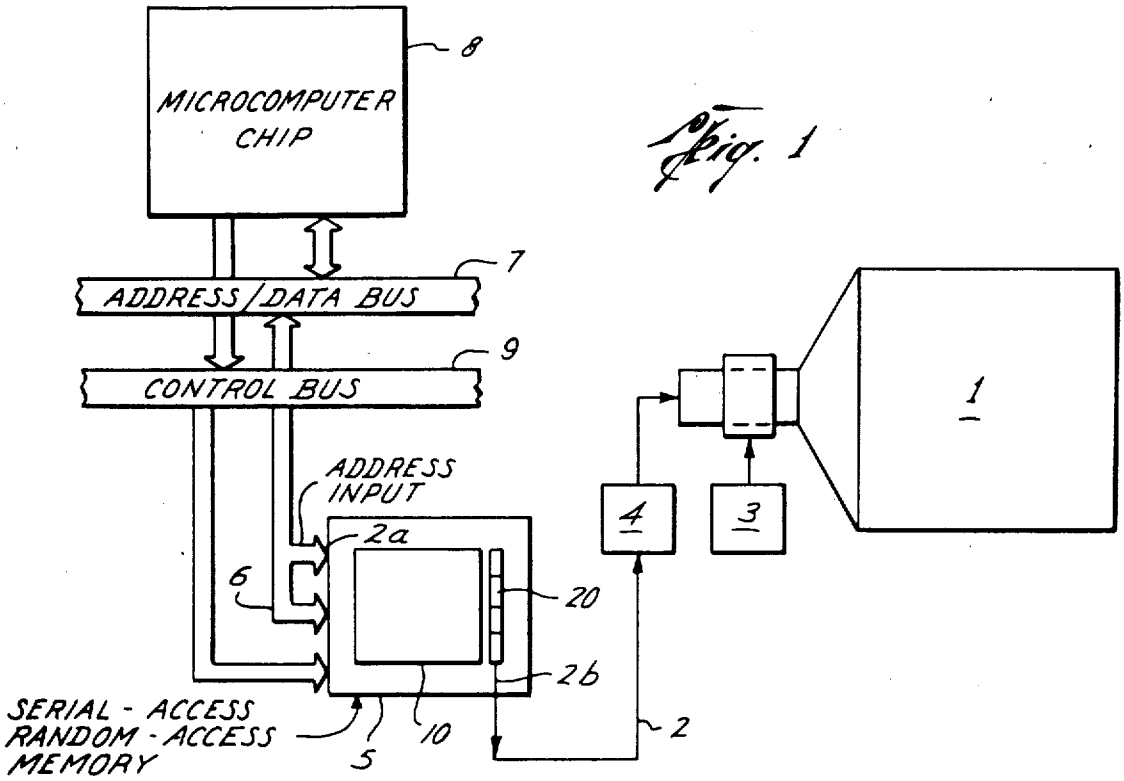

MICROCOMPUTER CHIP
8
ADDRESS/DATA BUS
7
CONTROL BUS
9
ADDRESS INPUT
2a
6
20
2b
SERIAL - ACCESS RANDOM - ACCESS MEMORY
5
10
2
4
3
1

Fig. 1

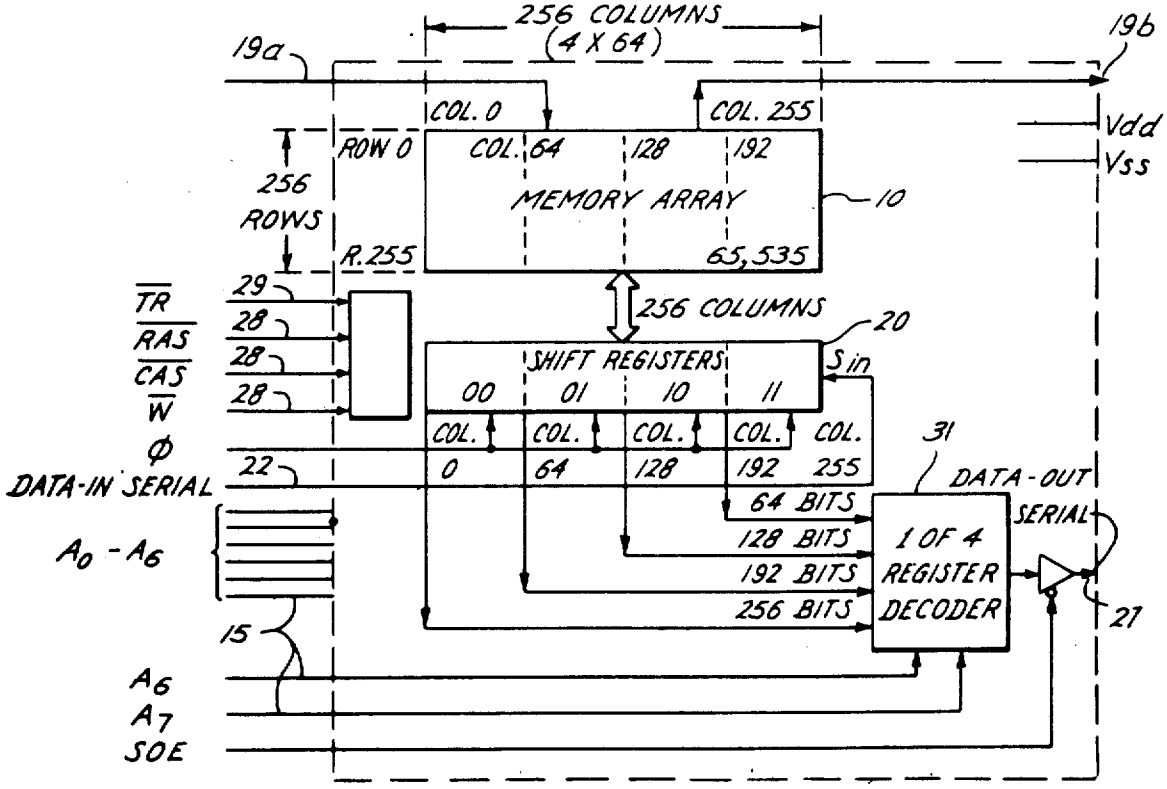

256 COLUMNS (4 X 64)
19a
19b
COL. 0
COL. 255
ROW 0
COL. 64
128
192
MEMORY ARRAY
10
256 ROWS
R.255
65,535
Vdd
Vss
TR
RAS
CAS
W
Φ
DATA-IN SERIAL
29
28
28
28
22
256 COLUMNS
20
SHIFT REGISTERS
S in
00
01
10
11
COL. 0
COL. 64
COL. 128
COL. 192
COL. 255
31
64 BITS
128 BITS
192 BITS
256 BITS
1 OF 4 REGISTER DECODER
DATA-OUT SERIAL
27
A0 - A6
15
A6
A7
SOE

Fig. 3

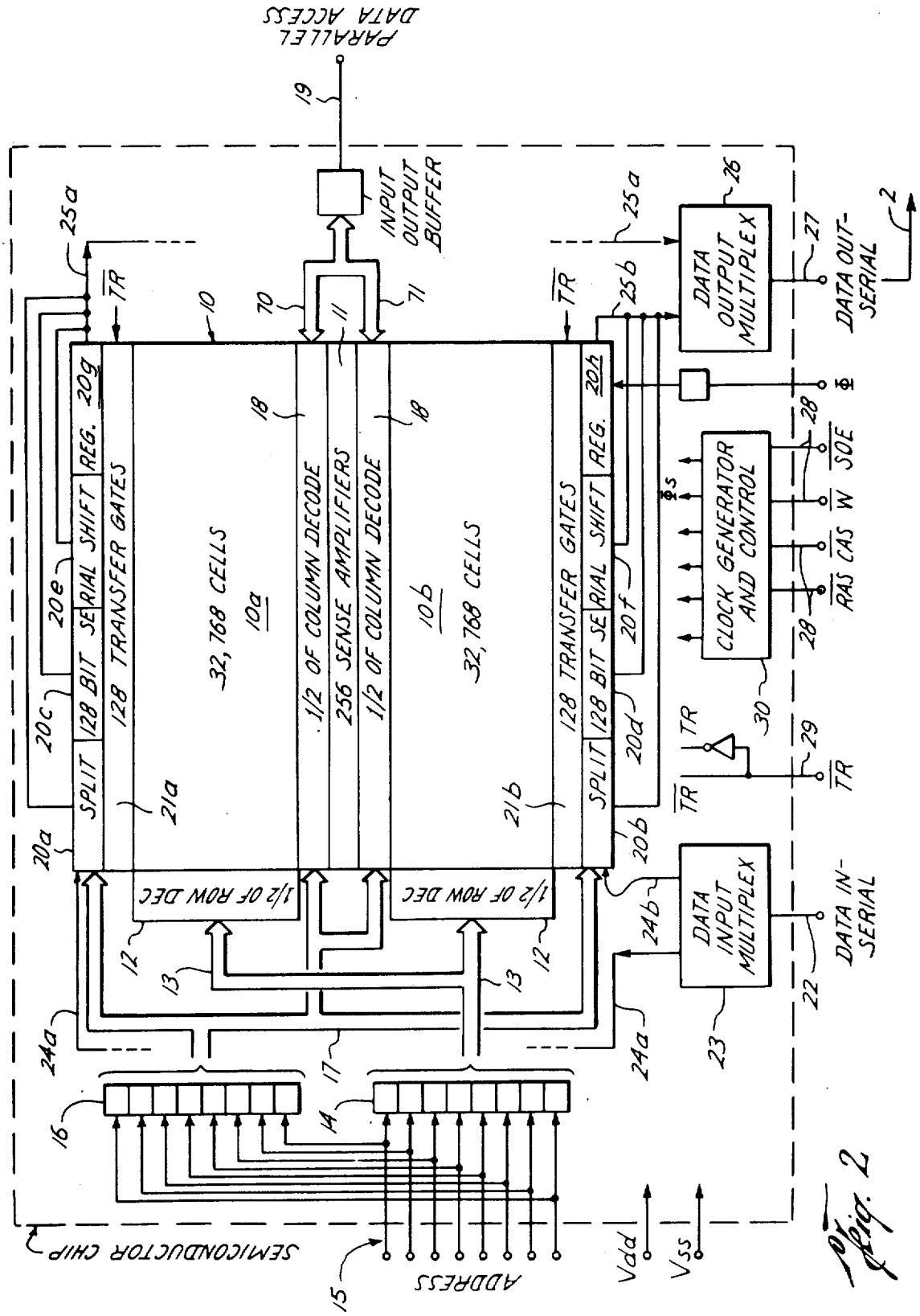

PARALLEL DATA ACCESS
INPUT OUTPUT BUFFER
DATA OUTPUT MULTIPLEX
DATA OUT-SERIAL
CLOCK GENERATOR AND CONTROL
DATA INPUT MULTIPLEX
DATA IN-SERIAL
SPLIT
128 BIT SERIAL SHIFT REG.
128 TRANSFER GATES
32,768 CELLS
1/2 OF COLUMN DECODE
256 SENSE AMPLIFIERS
1/2 OF ROW DEC
ADDRESS
SEMICONDUCTOR CHIP
Vdd
Vss
RAS
CAS
W
SOE
TR

Fig. 2

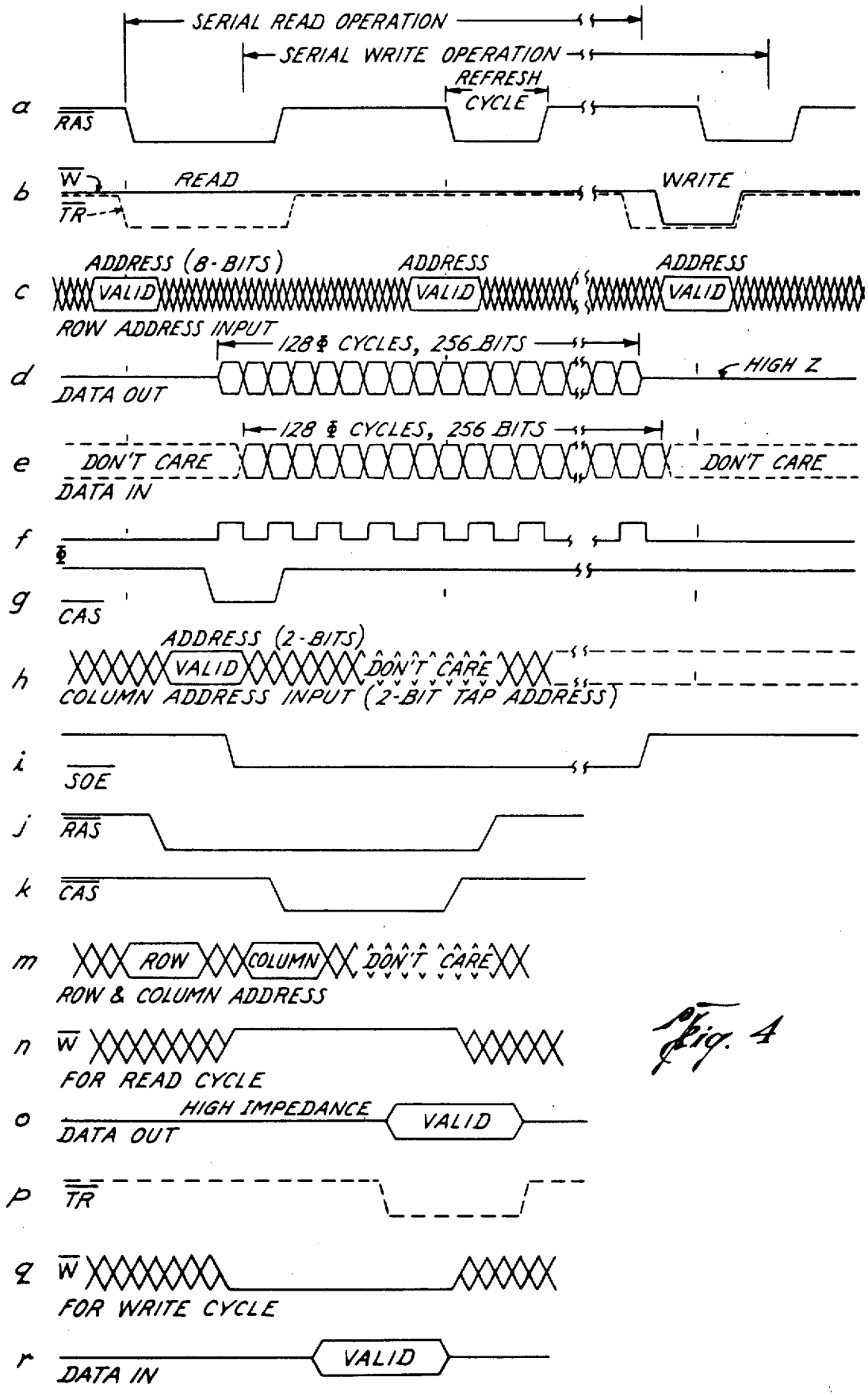

SERIAL READ OPERATION
SERIAL WRITE OPERATION
REFRESH CYCLE
a
RAS
b
W
TR
READ
WRITE
c
ADDRESS (8-BITS)
ADDRESS
ADDRESS
VALID
VALID
VALID
ROW ADDRESS INPUT
128 Φ CYCLES, 256 BITS
d
DATA OUT
HIGH Z
128 Φ CYCLES, 256 BITS
e
DON'T CARE
DON'T CARE
DATA IN
f
Φ
g
CAS
ADDRESS (2-BITS)
h
VALID
DON'T CARE
COLUMN ADDRESS INPUT (2-BIT TAP ADDRESS)
i
SOE
j
RAS
k
CAS
m
ROW
COLUMN
DON'T CARE
ROW & COLUMN ADDRESS
n
W
FOR READ CYCLE
o
HIGH IMPEDANCE
VALID
DATA OUT
p
TR
q
W
FOR WRITE CYCLE
r
VALID
DATA IN

Fig. 4

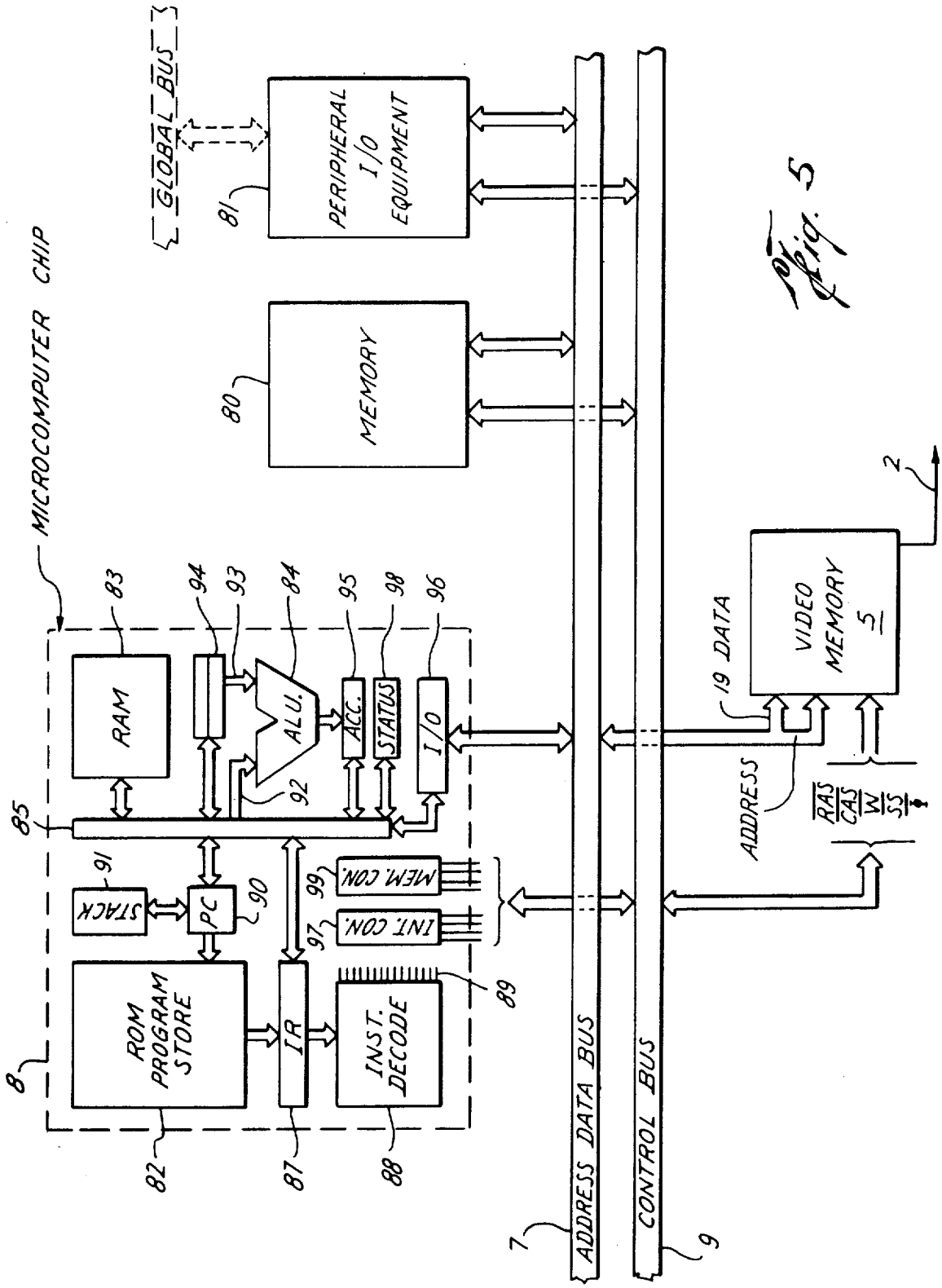

MICROCOMPUTER CHIP
GLOBAL BUS
PERIPHERAL I/O EQUIPMENT
MEMORY
RAM
ALU.
ACC.
STATUS
I/O
STACK
PC
ROM PROGRAM STORE
IR
INST. DECODE
MEM. CON.
INT. CON.
VIDEO MEMORY 5
19 DATA
ADDRESS
RAS
CAS
W
SS
ADDRESS DATA BUS
CONTROL BUS
8
7
9
2
80
81
82
83
84
85
87
88
89
90
91
92
93
94
95
96
97
98
99

Fig. 5

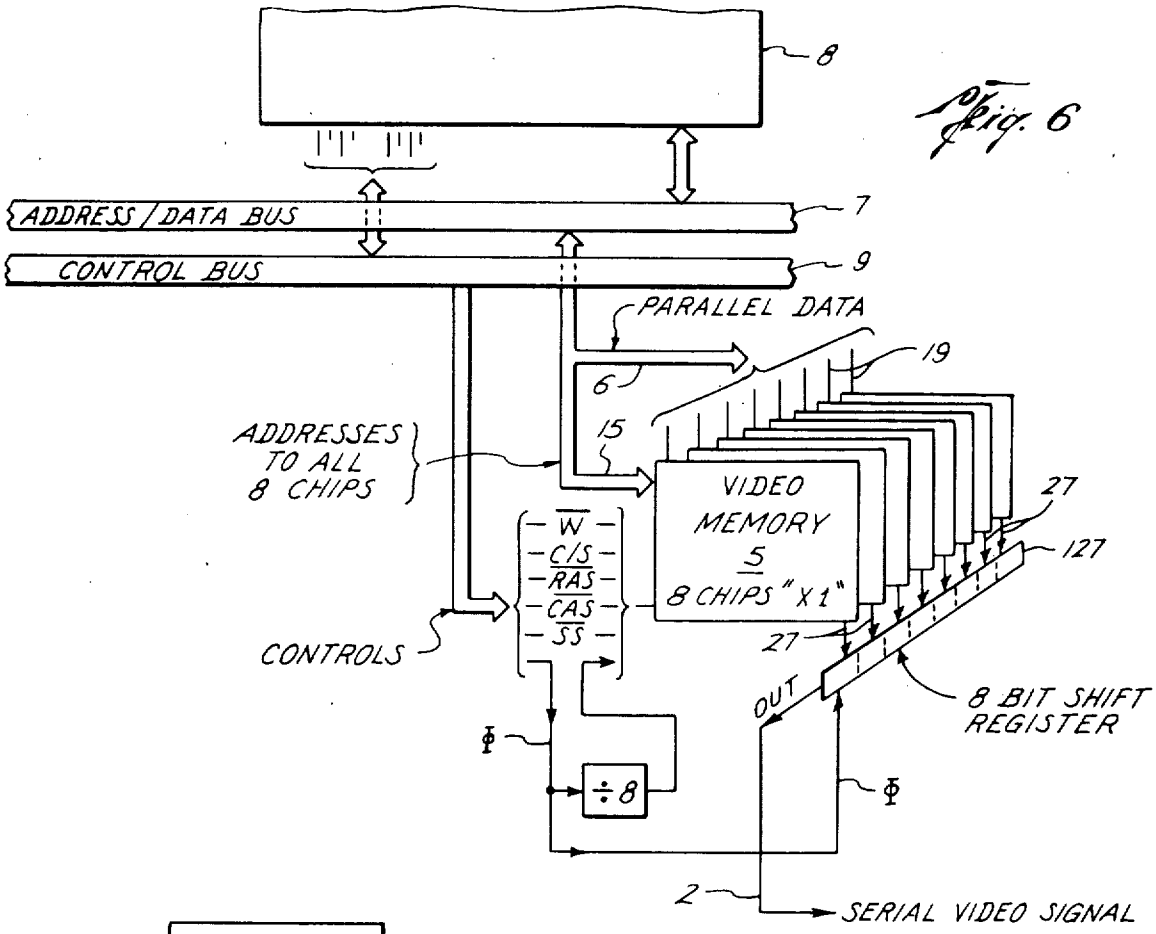

8
ADDRESS / DATA BUS
7
CONTROL BUS
9
PARALLEL DATA
19
6
15
ADDRESSES TO ALL 8 CHIPS
VIDEO MEMORY 5 8 CHIPS "X1"
27
127
W
C/S
RAS
CAS
SS
CONTROLS
27
OUT
8 BIT SHIFT REGISTER
Φ
÷8
Φ
2
SERIAL VIDEO SIGNAL

Fig. 6

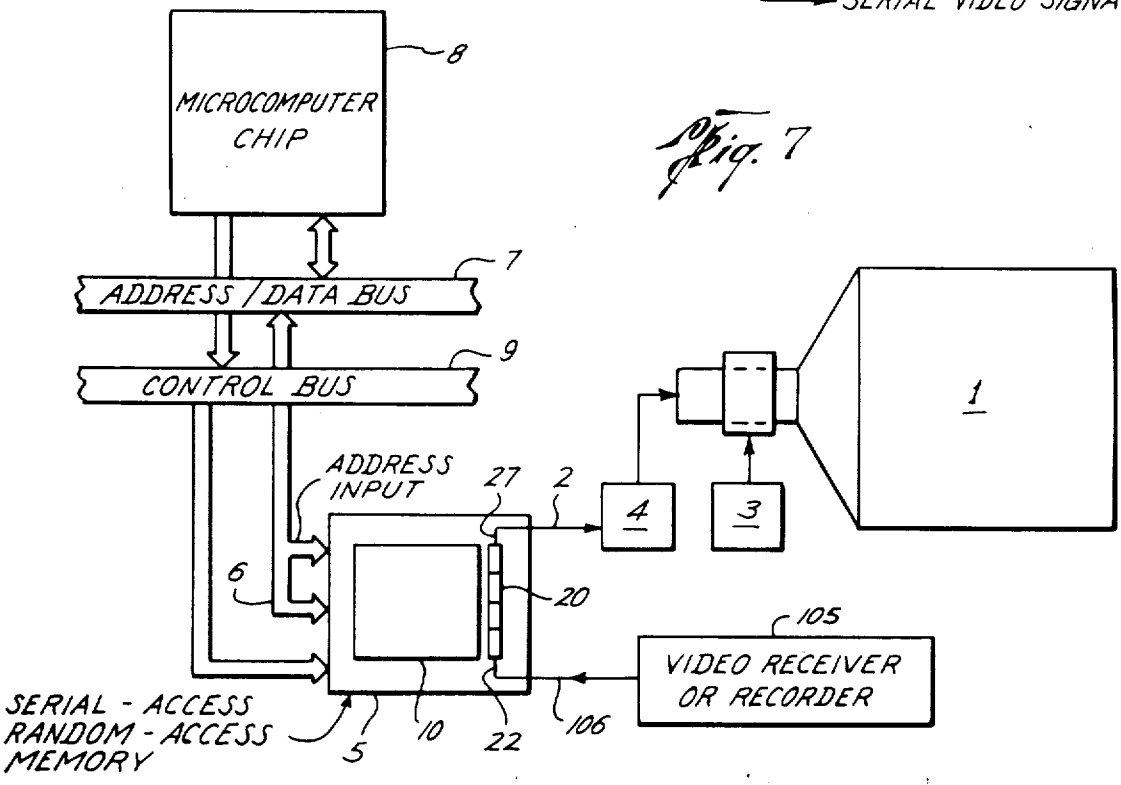

MICROCOMPUTER CHIP
8
ADDRESS / DATA BUS
7
CONTROL BUS
9
ADDRESS INPUT
27
2
4
3
1
6
20
105
VIDEO RECEIVER OR RECORDER
SERIAL - ACCESS RANDOM - ACCESS MEMORY
5
10
22
106

Fig. 7

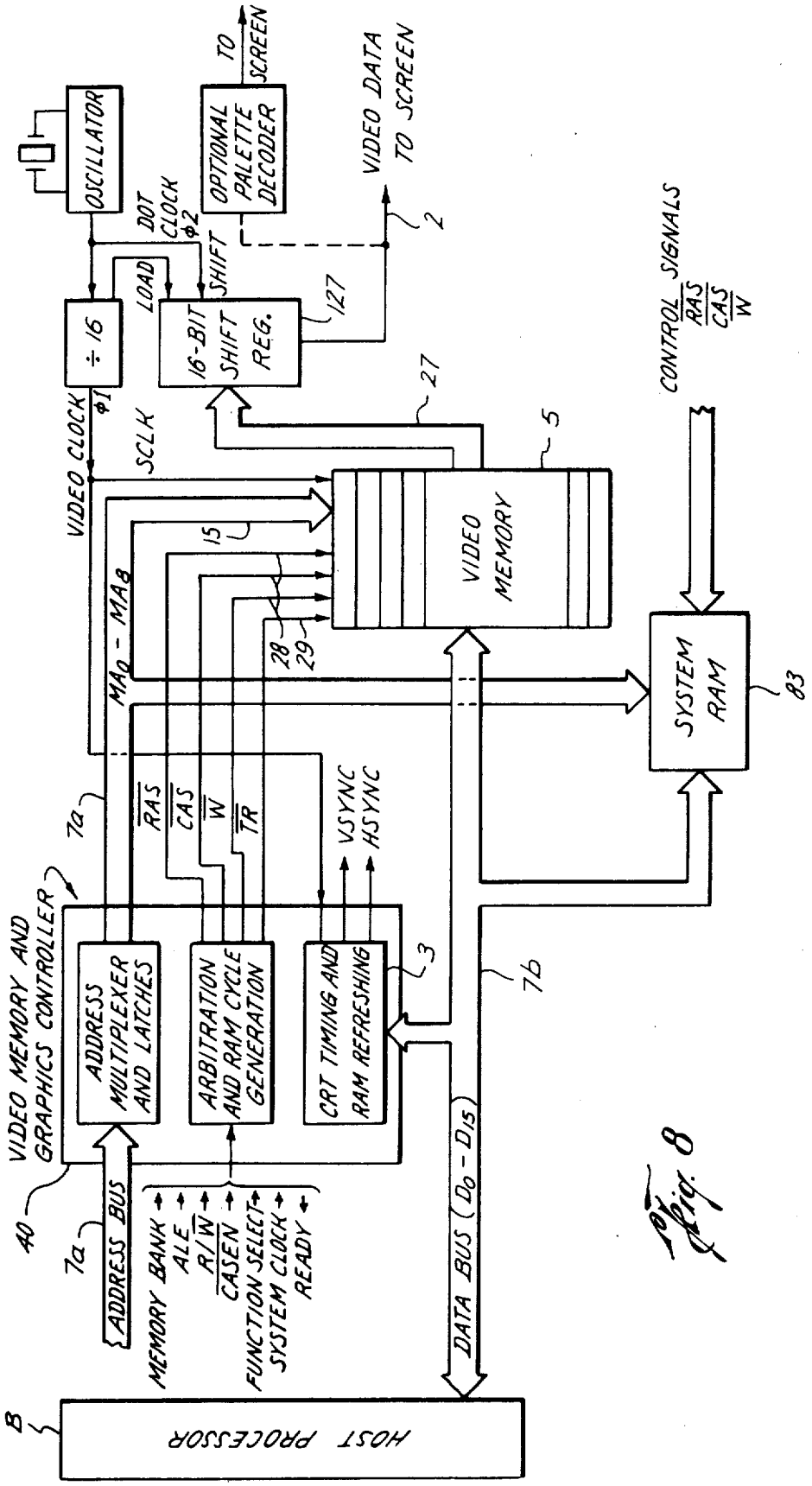

OSCILLATOR
÷ 16
VIDEO CLOCK
φ1
SCLK
DOT CLOCK
φ2
LOAD
SHIFT
16-BIT SHIFT REG.
127
OPTIONAL PALETTE DECODER
TO SCREEN
VIDEO DATA TO SCREEN
2
27
5
VIDEO MEMORY
15
28
29
MA0 - MA8
CONTROL SIGNALS
RAS
CAS
W
SYSTEM RAM
83
7a
RAS
CAS
W
TR
VSYNC
HSYNC
VIDEO MEMORY AND GRAPHICS CONTROLLER
ADDRESS MULTIPLEXER AND LATCHES
ARBITRATION AND RAM CYCLE GENERATION
CRT TIMING AND RAM REFRESHING
3
7b
A0
7a
ADDRESS BUS
MEMORY BANK
ALE
R/W
CASEN
FUNCTION SELECT
SYSTEM CLOCK
READY
DATA BUS (D0 - D15)
B
HOST PROCESSOR

Fig. 8

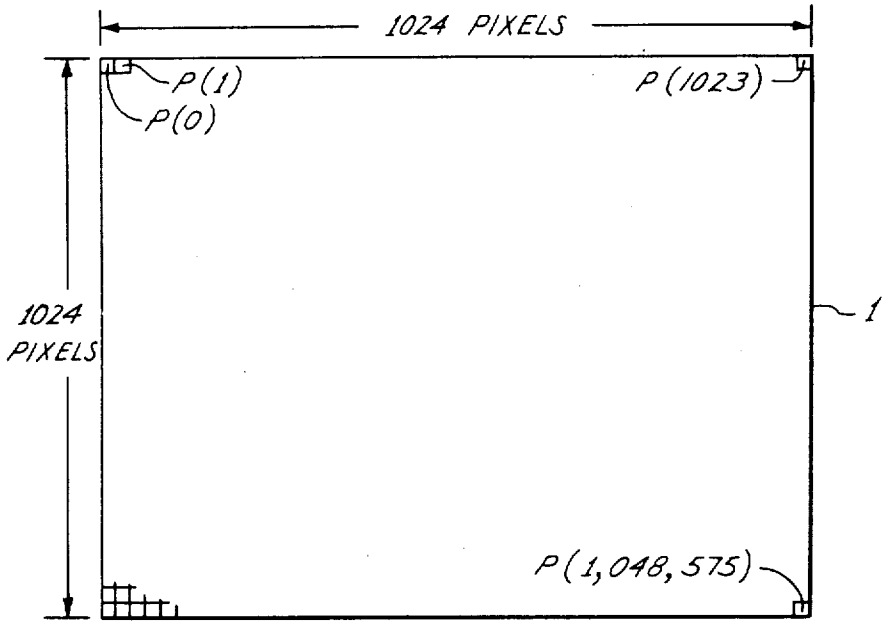

1024 PIXELS
P(1)
P(0)
P(1023)
1024
PIXELS
1
P(1,048,575)

Fig. 9

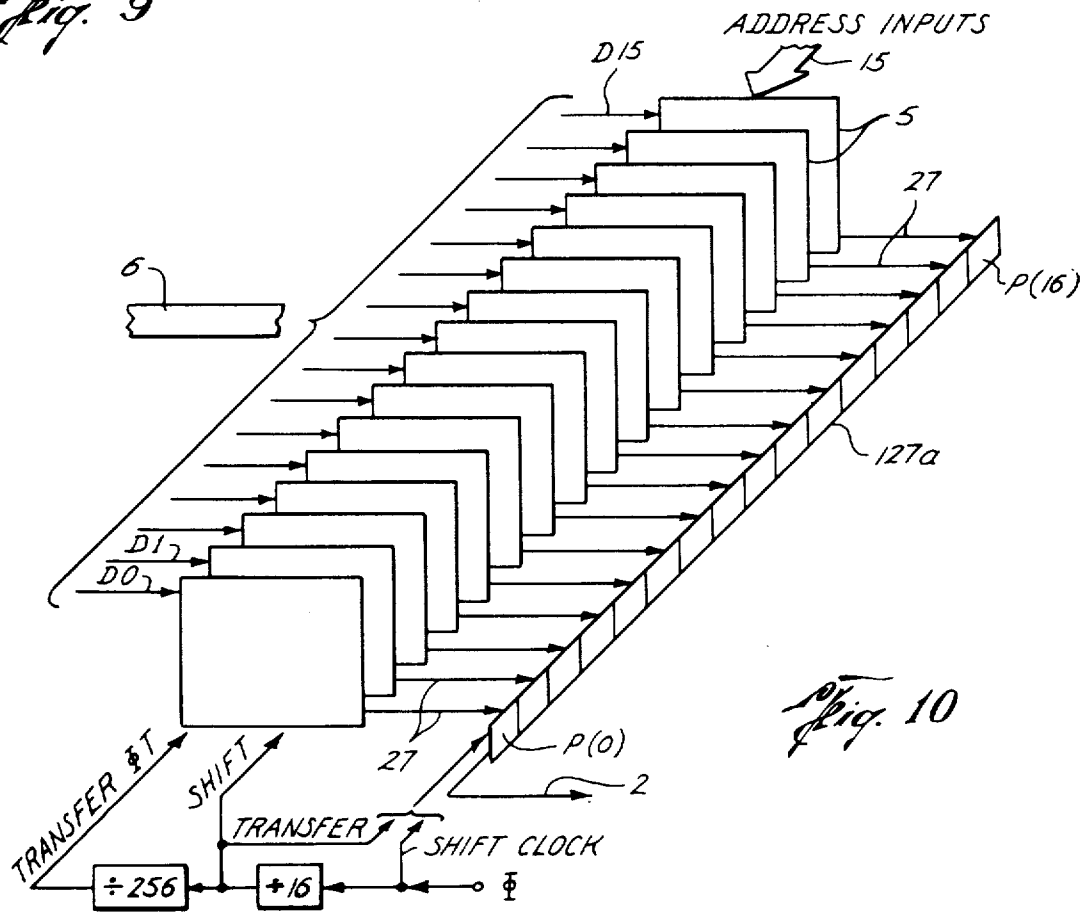

ADDRESS INPUTS
D15
15
5
27
6
P(16)
127a
D1
D0
27
P(0)
2
TRANSFER $T
SHIFT
TRANSFER
SHIFT CLOCK
÷256
÷16
Φ

Fig. 10

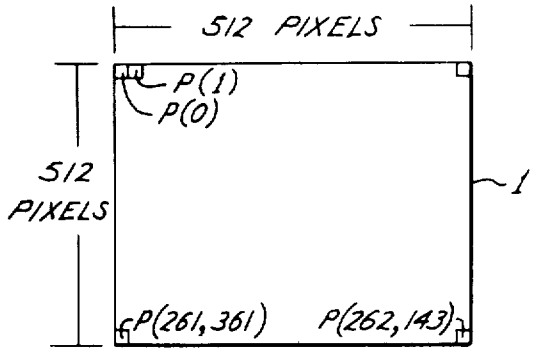


Fig. 11

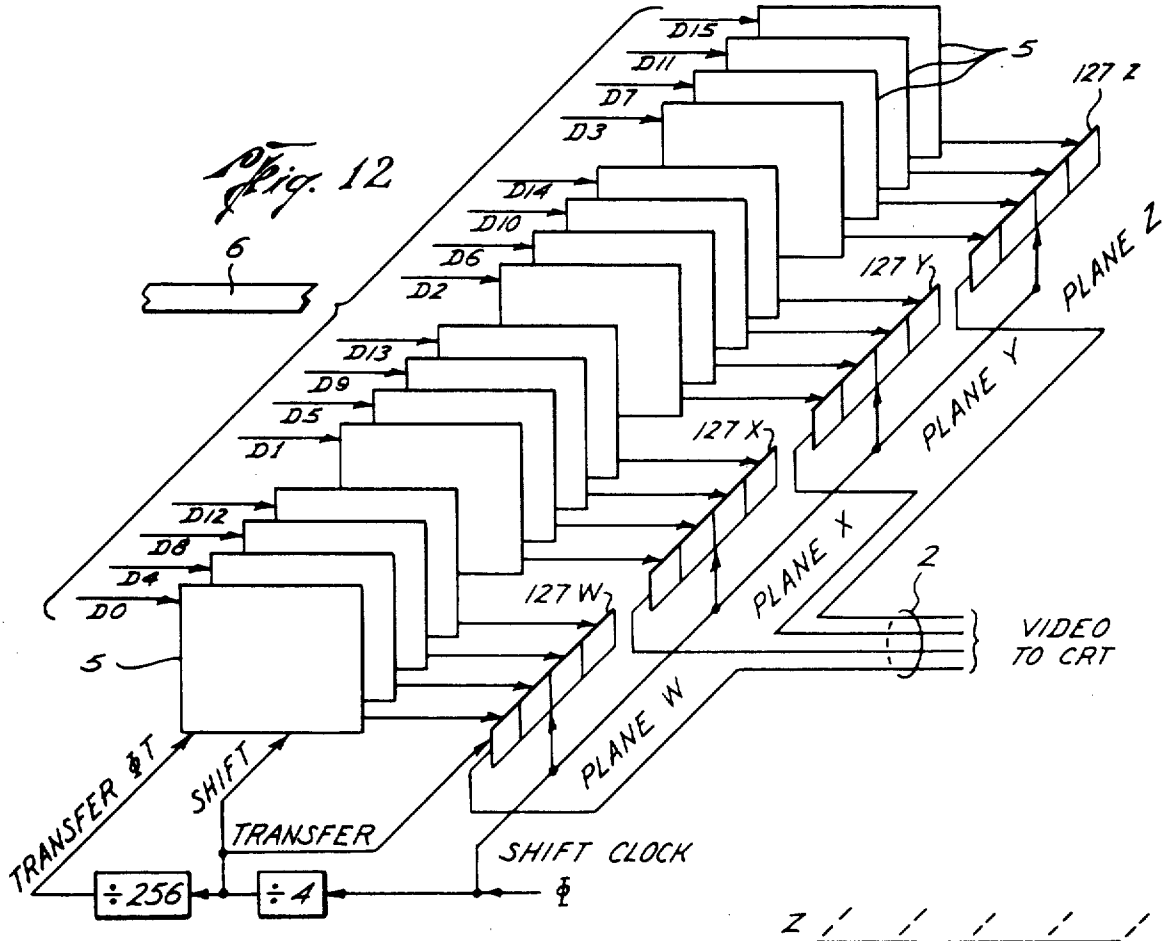


Fig. 12

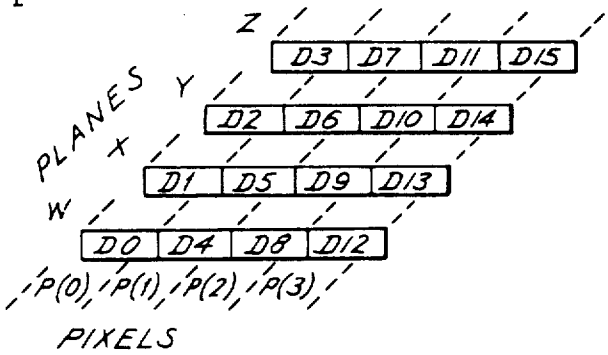


Fig. 13

RANDOM/SERIAL ACCESS MODE SELECTION CIRCUIT FOR A VIDEO MEMORY SYSTEM

BACKGROUND OF THE INVENTION

This invention relates to improved video-type computer systems and the like, and more particularly relates to a simplified circuit for recovering stored video data signals.

It is well known that video-type displays are preferred for computer games and so-called "home" computers, and that there is an increasing demand in this market for systems which exhibit more complex video images. It is a relatively simple task to produce a video image having increased resolution, provided that cost is not a factor. However, the home computer market is characterized by stiff price resistance above a certain level, and therefore there is a need for new and simpler designs to achieve this objective.

The problem is basically concerned with the storage circuit which holds the video signals or data sought to be presented on the CRT screen. It is well known that the complexity of the video image is a function of the number of pixels on the CRT screen, and it is further well known that the associated storage circuit must employ at least as many memory cells as there are pixels on the CRT screen. Accordingly, if the number of pixels is sought to be increased for this purpose, then a corresponding increase must be made in the number of memory cells in the storage circuit. Unfortunately, this results in an increase in the cost of the storage circuit which is disproportionate to the improvement in the resolution of the image, not only because of the need for more memory cells in the system, but also because of a need for more complex circuits for recovering data signals stored in the memory circuit.

The problems, which attend an increase in the number of memory cells, also relate to an increase in defective assemblies. This is because most storage circuits employ "RAM chips" of conventional design, and these components are conventionally fabricated with pin-type input and output terminals. A pin-type terminal is difficult to fabricate without defects, and if the chip is required to have an increased number of these terminals, it is clear that an increase in the number of pin-type terminals will necessarily increase the chances of a defect in the individual memory chips. Accordingly, it will further be apparent that if it is necessary to increase the number of memory cells in the storage circuit, this will increase the manufacturing cost of the overall system.

These disadvantages in the prior art are overcome or mitigated with the present invention, however, and simpler and more reliable storage circuitry and technique are herewith provided.

A video display system employing a bit-mapped dual-ported video memory having both serial and parallel access is disclosed in U.S. patent application Ser. No. 427,236, filed Sept. 29, 1982 (now U.S. Pat. No. 4,562,435) by McDonough, Laffitte and Hughes, assigned to Texas Instruments.

SUMMARY OF THE INVENTION

In a preferred embodiment of the present invention, a video-type computer system is provided which employs a microprocessor for generating video data signals as required, a CRT circuit for generating a video image as a function of these signals, and an appropriate storage

circuit therebetween. As will hereinafter be explained in detail, the processor is preferably programmed to generate appropriate command and address signals for causing the storage circuit to yield up its stored data signals as required. In addition, therefore, two or more signal or data buses are appropriately included for transmitting these signals to the memory unit, and for delivering stored data signals back to the microprocessor circuit.

It should be noted that the storage circuit preferably employs a RAM unit or chip. In many prior art computer systems, this RAM chip is connected to yield up stored data whenever it receives a "read" signal of the type which is generated by the aforementioned microprocessor (or its functional equivalent). Although the microprocessor is both the source of this "read" signal and the intended recipient of the data sought to be recovered, it sometimes happened, in these prior art systems, that the output video data from this RAM unit will appear at times which are inconvenient to the microprocessor. Furthermore, this is especially a problem with those systems wherein the number of memory cells has been increased by means of additional RAM chips, and where each of these RAM chips is arranged to store and deliver output signals.

In some prior art systems, this particular problem is met by arranging for the storage circuit to deliver up stored data only in response to an output data control signal generated by the microprocessor along with the conventional "read" signal. Although this expedient eliminates the problem of generating data signals at inopportune times, it also calls for additional circuitry in the system including both an additional bus and an additional pin in each of the RAM chips. The need to add this circuitry is, of course, inconsistent with the objective of keeping the ultimate cost of the product below the level at which price resistance is expected to occur.

In the present invention, this is avoided by arranging the storage circuit so that the data command signal will perform both of two separate different functions, whereby the need for additional bus and other circuitry, and whereby the need for the additional pin in the RAM chip is eliminated. Even if a memory component is selected which already has the extra pin, there is nevertheless an advantage is not being required to interconnect with this unused pin, inasmuch as each pin is a potential source of failure in the system.

The advantages of the present invention are best achieved by generating and using these signals in different sequences, depending upon the function sought to be performed. More particularly, the storage circuit is arranged so that when the data output control signal is generated concurrently with the row address signal, the RAM unit will operate in the serial mode and stored data can thereafter be transferred to its serial output pin in response to the "read" signal from the microprocessor. Alternatively, the RAM unit may be put into its parallel mode by the occurrence of the row and column address signals, and then the "read" signal, whereby the data output control signal is then employed to transfer stored data to the parallel output pin as a function of the "read" signal from the microprocessor.

Accordingly, it is an object of the present invention to provide a simplified storage circuit for handling the storage and recovery of video data in a ccomputer system and the like.

It is also an object of the present invention to provide an improved storage circuit, which is simpler and cheaper to manufacture, and which is comparatively freer of faults and defects.

It is further a specific object of the present invention to provide a method of recovering data from a RAM circuit having parallel and serial outputs, comprising, generating a data output control signal to said circuit, generating a row address signal to said circuit, and delivering stored data to said outputs as a function of the sequence of said data output control and said row address signals.

It is also a specific object of the present invention to provide an improved video circuit comprising, processor means arranged and adapted to generate row and column address signals and first and second control signals, control bus means interconnected with said processor means for carrying said control signals, data bus means interconnected with said processor means for carrying said address signals, and memory means interconnected with said buses and operable in a parallel mode in functional response to a first sequence of said control signals and in a serial mode in functional response to a second sequence of said control signals.

These and other objects and features of the present invention will become apparent from the following detailed description, wherein reference is made to the figures in the accompanying drawings.

IN THE DRAWINGS

FIG. **1** is a simplified block diagram of a computer system embodying one form of the present invention.

FIG. **2** is a more detailed block diagram of a selected portion of the computer system depicted in FIG. **1**.

FIG. **3** is another different block diagram of the apparatus depicted FIG. **2**.

FIGS. **4***a*–**4***r* are simplified time-dependent represenations of certain signals generated for operation of the apparatus depicted in FIG. **2**.

FIG. **5** is a block diagram of another portion portion of the system depicted in FIG. **1**.

FIG. **6** is a block diagram of a further different portion of the system depicted in FIG. **1**.

FIG. **7** is a block diagram of another different embodiment of the system depicted in FIG. **1**.

FIG. **8** is a block diagram of a further different embodiment of the system depicted in FIG. **7**.

FIG. **9** is a simplified depiction of a video screen suitable for the purposes of the present invention and having 1024 by 1024 pixels.

FIG. **10** is a simplified functional diagram of a portion of the system depicted in FIG. **1** according to another embodiment of the invention.

FIG. **11** is a simplified depiction of a video screen suitable for the purposes of the present invention and having **512** by **512** pixels for color presentations.

FIG. **12** is functional diagram of a portion of the system depicted FIG. **1** according to another different embodiment of the invention.

FIG. **13** is a simplified representation of the correspondence between color planes and pixels for the purpose of the structures in FIG. **12**.

DETAILED DESCRIPTION OF SPECIFIC EMBODIMENT

Referring to FIG. **1**, there may be seen a simplified representation of the components of a computer video system which employs a dual-port bit-mapped memory arrangement according to one embodiment of the invention. More particularly, there may be seen a video display section **1** having a conventional raster-scanned CRT tube responsive to a video signal input **2** consisting of serial bits of data arriving at a rate of about 20 MHz or greater.

The standard TV signal normally provides 60 data frames per second, with 512 interlaced lines per frame, and each such line may have several hundred dots or pixel. The product of these numbers indicates that data bits must be supplied to the CRT at frequencies of the order of 20 MHz. For a simple black and white picture, each dot or pixel can be defined by one data bit, but up to four bits may be required for images in a more realistic sixteen shades of gray. A full-color image may require three or four streams or planes of the data and usually will require at least one byte (8-bits) per pixel.

The horizontal and vertical scanning and synchronizing circuitry **3** and video signal shaping circuitry **4** may be of a conventional design, and are not a functional part of the present invention. In addition, the circuitry depicted in FIG. **1** may also include a complete TV monitor or receiver as needed. However, the video data on input **2** is preferably received from a bit-mapped video memory **5** as will be described later herein, and this memory **5** will preferably have at least one cell for each corresponding pixel on the video screen **1**. The memory **5** may conveniently have a "parallel" or random input/output port **6**, or it may have separate parallel input and output ports, in addition to the serial input port **2***a* and serial output port **2***b* depicted in FIG. **1**. In addition, port **6** is preferably coupled to a multiplexed address/data input/output bus **7** of a suitable microcomputer or microprocessor **8**.

Referring again to FIG. **1**, it will be noted that the memory **5** receives addresses appearing on the bus **7**, to define the address for the serial ports **2***a* and **2***b*, and also to define addresses for writing into or reading from the memory **5**, by way of the parallel or random port **6**. A control bus **9** coupling the microprocessor **8** to the memory **5** provides the basic clock frequency signal ϕ which clocks the serial video data onto the CRT input line **2**, and which also provides the memory control signals such as Address Latch, Row Address Strobe ($\overline{RAS}$), Column Address Strobe ($\overline{CAS}$), Serial Select, Serial Output Enable (SOE), Write Enable, Write ($\overline{W}$), Increment (INC), etc., as may be required because of the characteristics of both the memory device **5** and the microcomputer **6**.

The memory **5** further includes a memory or storage array **10** composed of rows and columns of memory cells partitioned according to the size and type of video display **1** and the type of memory **5** which has been chosen. In addition, the memory **5** further includes a serial shift register **20** having a plurality of taps. In this regard, it will be noted that a standard two-level black and white TV raster conventionally requires about 512 rows of cells (for 512 lines) arranged in 512 columns (for 512 pixels per line), to accomodate a total of 256k-bits of memory per complete data frame. Accordingly, if 64k memory devices are used for these purposes, then four such units will be required to form the memory **5**, or else one 256k-bit memory device may be used instead. These four units may alternate in feeding 256-bit blocks of video data serially onto the line **2**, or other formats may be employed as appropriate. A black and white image having relatively low resolution may be pro-

duced using only one 64k-bit storage array capable of providing 256 lines by 256 pixels per line.

One example of a memory device **5** which may be suitable for use in the system depicted in FIG. **1** and depicted in FIG. **2**, is a 64k-bit MOS dynamic read/-write memory using one transistor cells, as shown in U.S. Pat. No. 4,239,993, and further including a serial shift register having multiple taps added. For this example, the random access may be one bit wide. Other suitable examples (not shown) may be memory devices as hereinbefore described which have 256k-bits of storage or even larger.

As hereinafter set forth, if the memory is partitioned to provide eight chips, for example, then the individual storage devices may be X1, i.e. one bit wide, and eight of these storages may be connected in parallel for access by a typical 8-bit microcomputer 8. Other partitioning, such as X4 or X8, could also be employed as will hereinafter be apparent.

The memory device **5** depicted in FIG. **2** is typically made by an N-channel, self-aligned, silicon-gate, double-level polysilicon, MOS process, with all of the device being included in one silicon chip of about 1/30 of a square inch in size, which usually would be mounted in a standard dual-in-line package having twenty pins or terminals. For a 256k-bit device this package may be provided with as many as twenty-two pins or terminals. Similarly, the number of the pins would increase for larger volume devices. The device includes in this example an array **10** split into two halves **10***a* and **10***b* of 32,768 cells each, in a regular pattern of 256 rows and 256 columns. Of the 256 rows or X lines, there are 128 in the array half **10***a* and 128 in the half **10***b*. The 256 columns or Y lines are each split in half with one-half being in each of the halves **10***a* and **10***b*. There are 256 sense amplifiers **11** in the center of the array; these are differential type bistable circuits made according to the invention disclosed and claimed in said U.S. Pat. Nos. 4,239,993, or in 4,081,701. Each sense amplifier is connected in the center of a column line, so 128 memory cells are connected to each side of each sense amplifier by a column line half. The chip requires only a single 5 V supply Vdd, along with a ground terminal Vss.

A row or X address decoder **12**, split into two halves, is connected by sixteen lines **13** to eight address buffers or latches **14**. The buffers **14** are made according to the invention disclosed in U.S. Pat. No. 4,288,706. An eight-bit X address is applied to inputs of the address buffers **14** by eight address input terminals **15**. The X decoder **12** functions to select one of the 256 row lines as defined by an eight bit address on the input terminals **15** recieved via bus **7** from the microcomputer **8**. For more than 256 row lines, i.e. a 256k-bit memory with 512 row lines, a larger than eight-bit X address and eight-bit latch must be employed.

A column address is also received on the input pins **15** and latched into column address latches **16**. For a bit-wide random-access data input/output, all eight column address bits are needed, but for byte-wide access, i.e. eight bits, only five address bits are needed, and the microcomputer may output additional column address bits to select among several cascaded chips; these additional column address bits may be used by chip-select decoders of conventional construction. The outputs of the column address latches **16** are connected by lines **17** to a decoder **18** in the center of the array which selects one-of-256 columns to produce a bit wide input/output on random access input/output line **19**; separate input **19***a* and output **19***b* lines may be used as shown in FIG. **3**, or the lines **19** may be multiplexed as shown in FIG. **2**. Rows of dummy cells (not shown) are included on each side of the sense amplifiers as is the usual practice in devices of this type. As for the X-address, for larger volume devices, the number of bits and latches required to identify a column increases.

The memory device is thus similar to a standard dynamic RAM, with bit-wide or other bit-size random access and also having a serial input/output. Continuing to refer to FIG. **2**, the serial access is provided by a 256k-bit serial shift register **20** split into two identical halves with the halves positioned at opposite sides of the array **10**. The same result may be achieved by placing both halves on the same side of the array, but laid out one above the other. However, placing the halves on opposite sides of the array balances the operation of the sense amplifiers.

The shift register **20** may be loaded from the column lines of the array **10** for a read cycle, or loaded into the column lines for a write cycle, by 128 transfer gates **21***a* on one side of the array and a like number of transfer gates **21***b* on the other side of the array.

Data input to the device for serial write is by a data-in terminal **22** (**2***a* of FIG. **1**) which is connected by a multiplex circuit **23** to inputs **24***a* and **24***b* of the shift register halves. Data is read out serially from the register halves via outputs **25***a* and **25***b*, a data-out multiplex and buffer circuit **26**, and a data-out terminal **27** (**2***b* of FIG. **1**).

The shift rgister **20** is operated by a clock ϕ which is used to shift the bits through the stages of the register, two stages for each clock cycle. For read operations it takes only 128 cycles of the clock ϕ to output 256 bits from the 256 bit positions of the split shift register. A control signal $\overline{TR}$ **29** applied to the transfer gates **21***a* and **21***b* connects each of the 256 bit positions of the shift register **20** to its corresponding column line in the array halves **10***a* and **10***b*. However, according to the invention, this same control signal $\overline{TR}$ **29** performs the additional function of controlling the availability of selected random data on the random output terminal **19***b* (See FIG. **3**), as discussed later herein. This allows for multiplexing both the data input and output busses with the address bus to reduce interconnecting circuitry in a video computer system and the like, and thereby reduce costs for such a system and improve its reliability.

In a serial write operation, the sense amplifiers **11** are operated by a write command, $\overline{W}$, occuring after $\overline{TR}$ to set the column lines at a full logic level, after which one row line is selected by the address in the latches **14** and the data forced into the memory cells of this row. A serial read cycle starts with an address on the input **15** which is decoded to activate one of the 256 X or row address lines (and a dummy cell on the opposite side). The sense amplifiers **11** are then actuated by a control signal from clock generator and control circuitry **30** to force the column lines to a full logic level, and then the transfer gates **21***a* and **21***b* are actuated by control signal $\overline{TR}$ to move the 256 bits from the selected row into the corresponding shift register **20** halves. The shift clock signal ϕ is then applied and may move 256 bits onto the output pin **27** in serial format via the multiplex circuit **26**, at two stages or bits per clock cycle, requiring 128 clock cycles for the entire register. The output pin **27** is connected to the video input **2** of FIG. **1**.

The memory device is thus similar to a standard dynamic RAM with a bit-wide or other bit-size random access with a serial input and output; the 256-bit serial shift register **20**, which provides the serial input and output, is organized as four 64-bit shift registers. One, two, three or four 64-bit shift registers may be accessed depending upon which of the four "taps" along the 256-bit shift register is selected. Since the 256-bit shift register is split into two "halves", each 64-bit shift register is also split into halves. As shown in FIG. **2**, one 64-bit shift register is top half **20***a* and bottom half **20***b*, a second 64-bit register is top half **20***c* and bottom half **20***d*, a third 64-bit shift register is top half **20***e* and bottom half **20***f*, and a fourth 64-bit shift register is top half **20***g* and bottom half **20***h*.

The tap selected determines whether one, two, three or all four 64-bit shift registers are accessed. The tap selected is determined by a two bit code applied to the two most signifcant column address inputs. The depiction in FIG. **2** is thus made of lines **17** from the column address latch **16** also inputting to the shift register **20** to select, via a binary code, the particular tap desired.

Referring now to FIG. **3**, the selection of the tap and the operation of the four 64-bit shift registers may be clearly understood. The memory array **10** is there seen to be divided into four 64 column by 256 row arrays. The four sets of 64 columns correspond to the four 64-bit shift registers. It may also be clearly seen that this memory device operates as if it were two devices, i.e., a 256 rows by 256 columns, random access array **10**, having an input **19***a* and output **19***b*, and an up to 256 by one array (i.e. the shift register **20**) having a serial input **22** and serial output **27**. This device allows access to the random array while serial data is being clocked into or out of one, two, three or four of the 64-bit shift registers. The register or tap decoder **31** of FIG. **3** may be of conventional construction. In this manner any bit of data may be accessed in 64 shifts or less, rather than 256 shifts. This may be of particular importance in other specific video applications.

Again, the tap selected is determined by a binary code, as illustrated in FIG. **3**, which is applied to the two most significant column address inputs **15**. These inputs feed into the tap or register decoder **31** as shown in FIG. **3**, to determine whether one, or two, or three or all four registers are accessed.

For example, if the two bits are both 0, then all 256 bits in the shift register may be shifted out. If the two bits are 01, then 192 bits, starting at bit 64, may be shifted out. If the two bits are 10, then 128 bits, starting at bit **128**, may be shifted out. The two bit code **11**, selects the last 64 bits starting at bit number **192** and then these last 64 bits may be shifted out. Similarly, the tap address bits may be used for shifting data into the selected portion of the shift register.

In general, the number of bit positions between any two adjacent taps is a power of 2 that is less than and a different power of 2 from that power of 2 which equals the number of bit positions in the shift register. Further, these taps may be used in 512 rows by 512 columns or 1024 rows by 1024 columns, or larger devices having corresponding 512-bit or 1024-bit or larger shift registers in the manner described hereinbefore. Larger numbers of taps than four taps, will correspondingly require a larger decoder and more inputs into the decoder via the most significant column address lines, and conversely for fewer than four taps.

By use of these "taps" along the shift register it is possible to manufacture a 1024 rows by 1024 column (1 M-bit) device that may be used with many different CRT screens, i.e., screens of various sizes (525 lines by 512 pixels per line, or 300 lines by 720 pixels per line), by merely employing the appropriate taps along the shift register of the device. Thus, all 1024 bits do not have to be shifted out of the register to obtain the desired data bits at the "end" of the register. Further, the taps could be located at bit positions of the register corresponding to the number of pixels per line in a "standard" CRT that does not correspond to a power of 2 position as described hereinabove.

Referring now to FIGS. **2** and **4**, the X or row address must appear on the inputs **15** when a row address strobe RAS, seen in FIG. **4***a*, is applied to a control input **28**. A column address strobe $\overline{CAS}$, as seen is FIG. **4***k*, and a read/write control $\overline{W}$ as seen in FIGS. **4***b*, **4***n* and **4***p* are other controls **28** that may be employed during random access to the device.

These external control inputs **28** are applied to clock generator and control circuitry **30** which generates a number of clocks and internal chip control signals to control the operation of various parts of the device. For example, when $\overline{RAS}$ goes low as seen in FIG. **4***a*, clocks derived from $\overline{RAS}$ cause the buffers **14** to accept and latch the eight row address bits then appearing on the input lines **15**. The row address must be valid during the time period shown in FIG. **4***c*, i.e. during the falling edge of $\overline{RAS}$. Serial access is one of the functions of the device controlled by a $\overline{TR}$ command on control input **29**. For a serial read operation, $\overline{TR}$ goes to active-low and the $\overline{W}$ signal is held high during the period seen in FIG. **4**b, and the data output on the serial output terminal **27** will occur during the time period of 128 cycles seen in FIG. **4***d*. For a serial write operation, the $\overline{TR}$ and $\overline{W}$ signal must both be active-low as also seen in FIG. **4***b* and the data-in bits must be valid during the preceding time period of 128 cycles seen in FIG. **4***e*.

Refresh occurs every time a row address appears on the inputs **15** and $\overline{RAS}$ goes low, as seen in FIG. **4***a* and **4***c*. Thus, during the 128 cycles when all four shift register halves are being read out through serial data-out pin **27**, refresh can be occurring by loading a new row address into the chip **5** along with a $\overline{RAS}$ signal. The shift register is not disturbed so long as $\overline{TR}$ does not occur in coincidence with a $\overline{RAS}$ signal as described hereinabove. Serial data can be shifted into the register halves while data is being shifted out, and so a write operation can begin just after a read operation is initiated; although not needed in the system of FIG. **1**, this feature is important for some other embodiment.

Parallel or random access occurs as illustrated in the timing diagram of FIGS. **4***j*–**4***r*; not that these Figures are on an expanded time scale compared to FIGS. **4***a*–**4***i*. The $\overline{X}$ or row address must appear on the inputs **15** when a row address strobe signal $\overline{RAS}$ is applied to an input **28**. Likewise, the Y or column address must appear during a column address strobe signal $\overline{CAS}$ on another input **28**. A read/write control $\overline{W}$ on an input **28** is the other control signal for the parallel or random access. When $\overline{RAS}$ goes low as seen in FIG. **4***j*, clocks derived from $\overline{RAS}$ cause the buffers **14** to accept and latch the eight row address TTL-level bits then appearing on the input lines **15**.

When $\overline{CAS}$ goes low as seen in FIG. **4***k* then clocks generated in the circuitry **30** cause the buffers **16** to latch the TTL-level column (or Y) address signals on

the inputs **15**. The row and column addresses must be valid during the time periods shown in FIG. **4***m*.

For a read cycle, the $\overline{W}$ signal on input **28** must be high during the period seen in FIG. **4***n*, and the output on the terminal **19** (see FIG. **4***o*) in response to the $\overline{TR}$ signal **29** being low as seen in FIG. **4***p*. Again, this allows one control signal $\overline{TR}$ **29** to first select the operation of the device as either serial or random, and second to control the application of the selected data to the random output terminal when the device is in the random mode. This allows one control pin to perform two control functions and thereby minimize the need for addition pins for the device. Minimizing pin count in turn reduces the cost of the device and improves the reliability of the device; reliability decreases and cost increases as the number of pins increase because of the need to make additional connections from the silicon chip to its lead case terminals. Reliability further decreases with increased numbers of pins because as the numbers of pins increase the silicon chip size increases, which increases the probability of manufacturing a defective chip from dust particles on the chip. For a write-only cycle, the $\overline{W}$ signal must be low as seen in FIG. **4***q* and the input must be valid on terminal **19** during the time seen in FIG. **4***r*.

The memory device also may operate in a page-mode format. That is, the row address is kept the same and successive column addresses are strobed onto the chip. By holding the row address the same, the time required to set up and strobe a new row address for the same page is eliminated and thus the time required for a page-mode operation is less than for a bit-mode format, as described hereinbefore. Typically, for the example device, a bit-mode operation requires about 260 nsec, where as a page-mode operation requires only about 160 nsec.

Further, the shift register **20** may be employed to rapidly read data into the array **10**. This may be accomplished by the microcomputer **8** generating the selected data bits and shifting these data bits serially into the shift register **20**. These data bits are then written into one of the rows selected by the microcomputer **8** using an eight-bit row address signal and the row address strobe ($\overline{RAS}$) control signal as discussed hereinbefore. The data bits are then written into a second row selected by the microcomputer **8**. This process of writing the data into rows continues until all of the rows contain the selected data bits.

This sequential writing of data from the shift register into all of the rows of the memory array is a particularly suitable process for initializing the data in the memory array to a known value (generally a "zero"). This process is useful for "powering up" the system to ensure only known data is read out of the array onto the CRT screen.

A modification of this process is for the microcomputer **8** to write the data into the individual cells of a selected row via a page-mode or bit-mode operation, and then transfer this data to the shift register. Once in the shift register it can then be sequentially written into the other rows as described hereinabove.

Either the modified or unmodified process allow for a much faster "clearing" or initialization of the memory array on "power-up", or as otherwise needed. This results from the writing of the data into a whole row of the array instead of the conventional process of writing the data into each individual cell, or each "cluster" of cells if in a byte or other bit-size system. The process is thus much faster than the conventional process for clearing the memory array.

The serial access via terminals **22** and **27** and shift register **20** is usually sequential in that the row address is incremented by one following each access. The video data is continuous stream of 256-bit serial blocks, one after the other, so the next address for serial access, after the memory to shift register transfer occurs, will generally be the last row address plus one. In the simplest embodiment, the microcomputer **8** sends out the row addresses for serial read, so an address counter in the microcomputer **8** will be incremented after each serial read is commanded. However, this function may also be performed by special circuitry on the chip of FIG. **2**. In contrast, the parallel access via terminal **19** is random rather than sequential and specific row and column addresses must be generated in the microcomputer **8**.

Thus, the timing of the $\overline{TR}$, $\overline{W}$, $\overline{CAS}$ and $\overline{RAS}$ signals is different for serial reads and writes, and random reads and writes. The voltages and timing of these control signals are seen in FIGS. **4***a*–**4***i* for serial operations and **4***j*–**4***r* for random operations. To load the shift register or transfer its contents into the array, $\overline{TR}$ must be low during the falling edge or $\overline{RAS}$. If $\overline{W}$ is held high during the falling edge of $\overline{RAS}$ then data in the columns of the selected row will be shifted into the register by the transfer gates **21**. The two bit tap address, see FIG. **4***h*, appearing on the most significant column address lines when $\overline{CAS}$ goes low, see FIG. **4***g*, determines which of the four cascaded 64-bit shift registers will be connected to serial output **27**.

The serial shift clock ϕ, FIG. **4***f*, then shifts the data out of the shift register at the desired data rate in response to the frequency of the clock ϕ. If $\overline{W}$ is held low during the falling edge of $\overline{RAS}$, then data is transferred from the shift register to the array columns and into the row selected by the row address bits. If $\overline{TR}$ is held high when $\overline{RAS}$ falls low then the shift register and array operate independent of each other i.e., the shift register may shift data out or in and the array is available for random access. Further, while in the random mode, the $\overline{TR}$ signal controls the timing of the appearance of the selected data on the random output terminal. This allows for multiplexing of the data and addresses busses to reduce costs and improve reliability of a computer video system.

Thus the timing of $\overline{W}$ is different in the random access mode and the serial mode of operation of the memory device. More specifically, $\overline{W}$ is set up and held relative to the falling edge of $\overline{CAS}$ during a random access operation. However, $\overline{W}$ is set up and held relative to the falling edge of $\overline{RAS}$ during a serial access operation.

Referring to FIG. **5**, a microcomputer which may be used with the system of the invention may include a single-chip microcomputer device **8** of conventional construction, along with additional off-chip program or data memory **80** (if needed), and various peripheral input/output devices **81**, all interconnected by an address/data bus **7**, and a control bus **9**.

A single bidirectional multiplexed address/data bus **7** is shown, but instead separate address and data busses may be used, and also the program addresses and data or input and output addresses may be separated on the external busses; the microcomputer may be of the Von Neumann architecture, or of the Harvard type or a combination of the two.

The microcomputer **8** could be one of the devices marketed by Texas Instruments under the part number

of TMS 7000, for example, or one of the devices commercially available under part numbers Motorola 6805, Zilog Z8 or Intel 8051, or the like. These devices, while varying in details of internal construction, generally include an on-chip ROM or read-only memory **82** for program storage, but also may have program addresses available off-chip, but in any event have off-chip data access for the memory **5**.

A typical microcomputer **8**, as illustrated in FIG. **5**, may contain a RAM or random access read/write memory **83** for data and address storage, an ALU **84** for executing arithmetic or logic operations, and an internal data and program bus arrangement **85** for transferring data and program addresses from one location to another (usually consisting of several separate busses). Instructions stored in the ROM **82** are loaded one at a time into an instruction register **87** from which an instruction is decoded in control circuitry **88** to produce controls **89** to define the microcomputer operation.

The ROM **82** is addressed by a program counter **90**, which may be self-incrementing or may be incremented by passing its contents through the ALU **84**. A stack **91** is included to store the contents of the program counter upon interrupt or subroutine. The ALU has two inputs **92** and **93**, one of which has one or more temporary storage registers **94** loaded from the data bus **85**.

An accumulator **95** receives the ALU output, and the accumulator output is connected by the bus **85** to its ultimate destination such as the RAM **83** or a data input/output register and buffer **96**. Interrupts are handled by an interrupt control **97** which has one or more off-chip connections via the control bus **9** for interrupt request, interrupt acknowledge, interrupt priority code, and the like, depending upon the complexity of the microcomputer device **8** and the system.

A reset input may also be treated as an interrupt. A status register **98** assoicated with the ALU **84** and the interrupt control **97** is included for temporarily storing status bits such as zero, carry, overflow, etc., from ALU operations; upon interrupt the status bits are saved in RAM **83** or in a stack for this purpose.

The memory addresses are coupled off-chip through the buffers **96** connected to the external bus **7** depending upon the particular system and its complexity. This path may be employed for addressing off-chip data or program memory **80** and input/output **81** in addition to off-chip video memory **5**. These addresses to bus **7** may originate in RAM **83**, accumulator **95** or instruction register **87**, as well as program counter **90**. A memory control circuit **99** generates (in response to control bits **89**), or responds to, the commands to or from the control bus **9** for address strobe, memory enable, write enable, hold, chip select, etc., as may be appropriate.

In operation, the microcomputer device **8** executes a program instruction in one or a sequence of machine cycles or state times. A machine cycle may be 200 nsec., for example, by an output from a 5 MHz crystal clock applied to the microcomputer chip. So, in successive machine cycles or states, the program counter **90** is incremented to produce a new address, this address is applied to the ROM **82** to produce an output to the instruction register **87** which is then decoded in the control circuitry **88** to generate a sequence of sets of microcode control bits **89** to implement the various steps needed for loading the bus **85** and the various registers **94**, **95**, **96**, **98**, etc.

For example, a typical ALU arithmetic or logic operation would include loading addresses (fields of the instruction word) from instruction register **87** via bus **85** to addressing circuitry for the RAM **83** (this may include only source address or both source and destination addresses). Such an operation may also include transferring the addressed data words from the RAM **83** to a temporary register **94** and/or to the input **92** of the ALU. Microcode bits **89** would define the ALU operation as one of the types available in the instruction set, such as add, subtract, compare, and, or, exclusive or, etc. The status register **98** is set dependent upon the data and ALU operation, and the ALU result is loaded into the accumulator **95**.

As another example, a data output instruction may include transferring a RAM address from a field in the instruction to the RAM **83** via bus **85**, transerring this addressed data from the RAM **83** via bus **85** to the output buffer **96** and thus out onto the external address/data bus **7**. Certain control outputs may be produced by memory control **99** on lines of the control bus **9** such as write enable, etc. The address for this data output could be an address on the bus **7** via buffer **96** in a previous cycle where it is latched in the memory **80** or memory **5** by an address strobe output from the memory control **99** to the control bus **9**.

An external memory controller device may be used to generate the $\overline{RAS}$ and $\overline{CAS}$ strobes. A two-byte address for the memory **5** would be applied to the bus **7** in two machine cycles if the bus **7** is 8-bit, or in one cycle if the bus is 16-bit.

The instruction set of the microcomputer **8** includes instructions for reading from or writing into video memory **5**, the additional memory **80** or the input/output ports of peripheral equipment **81**, with the internal source or destination being the RAM **83**, program counter **90**, temporary registers **94**, instruction register **87**, etc. In a microcoded processor each such operation involves a sequence of states during which addresses and data are transferred on internal bus **85** and external bus **7**.

Alternatively, the invention may use a microcomputer **8** of the non-microcoded type in which an instruction is excecuted in one machine state time. What is necessary in selecting the microcomputer **8** is that the data and addresses, and various memory controls, be available off-chip, and that the data-handling rate be adequate to generate and update the video data within the time constraints of the particular video application.

The video memory arrangement of the invention is described in terms of one bit data paths for the bus **7**, although it is understood that the microcomputer system and the memory technique is useful in either 8-bit or 16-bit systems, or other architectures such as 24-bit or 32-bit. One utility is in a small system of the type having 8-bit data paths and 12-bit to 16-bit addressing, in which no external memory **80** is needed and the peripheral circuitry **81** consists of merely a keyboard or like interface, plus perhaps a disc drive. A bus interface chip such as an IEEE 488 type of device could be included in the peripheral circuitry **81**, for example.

As illustrated in FIG. **6**, the video memory **5** may be confined as eight $\times$ 1 memory devices, or alternatively as one $\times$8 device. In this embodiment eight semiconductor chip memories **5** are used, all eight being 64K $\times$ 1 or perhaps 16K $\times$ 1, each having corresponding on-chip multiple shift registers for serial access as in FIG. **2**, with one bit wide input and output. For a full color television display **1**, using 8-bits per tri-color dot, a memory system consisting of four banks (eight chips per

bank) of 64k×1 memory devices would be required. Each line on the screen would use two 256-bit registers, clocked out one following the other, for each of eight video signal input lines 2 (instead of only one video data input 2 as shown).

The microprocessor 8 and bus 7 would access the 8-bit video data in parallel in a "×1" format on each chip by the eight data lines 6, one for each chip, as seen in FIG. 6. The address inputs 15 for all eight chips receive the same addresses from the bus 7, and all eight chips receive the same control inputs from bus 9. The eight serial outputs 27, one from each chip, are connected to respective bit positions of an eight-bit shift register 127.

The serial clock ϕ is divided by eight before application to the eight chips 5; the clock ϕ applied to the serial register 127 thus shifts out eight bits onto the video signal input line 2 and then another eight bits are loaded into register 127 from the shift registers 20 on the individual chips. Alternatively, instead of using the auxiliary shift register 127, the eight outputs 27 can be connected to eight parallel video signal inputs of the color TV.

For some video systems the serial data input 22 of FIG. 2 may be employed. The serial input may be video data from a receiver or a video tape playback mechanism 105 shown in FIG. 7 supplying a continuous serial video feed on line 106 to the input 22 of a chip as in FIG. 2. This incoming video data is written into the cell array 10 from the serial register 20, and while in the RAM array it is processed by the microcomputer 8 using the parallel access port 19, and then supplied to the video signal line 2 via the register 20 and the terminal 27.

An example of one use of this arrangement is to add text or graphics via the microcomputer on top of video supplied from the video receiver or tape recorder 105. Another example would be to enhance or correct the video from receiver or tape 105 by writing it serially into the array 10; reading the data out in parallel to store bits temporarily in the RAM 83 of the microcomputer, performing operations via the ALU 84, then writing the corrected data back into the array 20 via bus 7, from whence it it read out serially onto the video signal input 2. For this example the TR signal may be used to block the selected random data from the data bus until the microcomputer 8 is prepared to receive the data; this may become necessary when the microcomputer receives an interrupt signal in the midst of reading the video data.

The system allows for the desired portion of the register 20 to be serially loaded at the same time it is being serially read; that is, data-in and data-out overlap as seen in FIGS. 4*d* and 4*e*. During the 128, or fewer, clock cycles used for serial-in and serial-out, the array 10 can also be accessed in parallel by microcomputer 8 for the writeover, update or correction operation.

The semiconductor memory chip containing the array 10 may also include a conventional row address counter which generates an 8-bit 1-of-256 row address for coupling to the input of the row decoders 12 by multiplex circuitry, so the row decoder can accept an address from either the address input terminals 15 via buffers 14 or from the counter. This counter may be self-incrementing so that a count of one is added to the exisiting count whenever an input signal INC is received.

The counter may function as an onchip refresh address generator as set forth in U.S. Pat. Nos. 4,207,618 and 4,344,157, or as indicated in 4,333,167. A column address is not needed for refresh. When a row is addressed for serial-read or serial-write, this also refreshes the data in this row; likewise, a parallel access refreshes a row upon read or write. Thus, if the video data is being sampled via serial read at the usual rates needed for TV scan then each row is not addressed within the 4 msec refresh period (60 frames/second is about 17 msec between successive samplings).

During the time between serial reads, the microcomputer 8 will probably, but not necessarily, access all rows for parallel read or write often enough for refresh. Thus, the microcomputer program in the ROM 82 could include a counter loop to send out an incremented row address and RAS at some fixed rate to assure that the refresh address specifications are met. However, to avoid occupying the microcomputer program execution with refresh overload, the counter on the chip may provide the address on-chip, and the microcomputer 8 need only apply the $\overline{RAS}$ control. That is, upon receipt of $\overline{RAS}$ and no $\overline{CAS}$, with $\overline{W}$ and $\overline{TR}$ high, the multiplex is switched to apply the contents of the counter to the row decoder 12, and $\overline{W}$ is activated to refresh a row; no serial or paralllel data in or out is initiated.

An INC command (not shown) may be produced to increment the counter for the next refresh. Further, an on-chip refresh signal may be generated on-chip from a timer, as in U.S. Pat. No. 4,344,157, for example. The timer produces a refresh command at least once every (4 msec.)×(1/256)=16 microsec. This refresh command activates the input multiplexer, $\overline{W}$ and Inc just as the off-chip refresh request previously discussed.

The serial input and output via register 20, in most uses such as video, will require access to sequential rows. Thus, an on-chip 8-bit 1-of-256 counter may be employed to avoid the necessity of applying a row address from the microcomputer 8 for serial access. If the sampling rate is high enough, this may be the same as the refresh counter; i.e., only one counter is needed as no separate provision for refresh is necessary.

In particular, these on-chip counters may be used in the process of the invention to allow the microcomputer 8 to conduct other operations than "clearing" the memory array upon "power-up". That is, the microcomputer 8 need only shift the data into the shift register and then strobe the chip 5 with $\overline{RAS}$, $\overline{TR}$, $\overline{W}$ and INC signals to allow the on-chip counters to sequentially address each row in the array, to minimize occupying the microcomputer program execution with clearing overload.

Referring now to FIG. 8, a relatively complete video grahics subsystem employing the video memory 5 of the present invention, and a memory and graphics controller 40 is depicted. The memory and graphics controller would replace the conventional separate controllers needed for the video and main system memories and it may also be designed to generate the synchronization and blanking signals that are required by the display circuitry.

Each horizontal display scan is made up of an active display scan, during which data is sent out, and a retrace scan, when no data is required. During the retrace scan the shift registers 20 of the present invention may be reloaded from the memory array 10. The memory and graphics controller 40 could be designed with counters to keep track of the pixel and line position on the screen

so that the proper address and reloading information can be sent to the device automatically during the retrace period. This controller could also be designed with enough intelligence to handle interlaced or non-interlaced displays.

The use of a device employing the reduced pin count of this invention with such a graphics controller reduces the amount of external circuitry needed to control the video memory **5** and simplifies the signals the host processor **8** must generate to control the video display. Further, use of larger volume storage devices allows the use of a "standard" package of a graphics controller and video memory of use for CRT screens of varying sizes and resolutions.

The graphics controller **40** could also be designed to "clear" the video memory upon "power-up" by generating each row address, after receiving the "clearing" data bits from the host processor **8**, or could employ on-chip counters as described hereinbefore.

The shift clock ϕ may be generated separate from the microcomputer **8**. Note that serial access and parallel access to the array **10** via register **20** and line **19** may be asynchronous; that is, the ϕ generator need not be synchronized with the clock of the microcomputer **8**, but instead may be synchronized with the video display **1** of FIG. **1** or the video signal **106** from receiver **105** of FIG. **7**.

A system that advantageously utilizies these features is the embodiment of FIG. **7**, with serial input is an interactive home TV adapted for games, education use or catalog ordering, as examples. That is, a video background is fed into serial input **22** from cable or VCR, and the user superimposes his input via microcomputer **8** (employing a keyboard, joystick, or the like coupled via input/output **81**), and the resulting composite video is applied to the screen **1** via line **2**. This same video data, or alternatively only the variable added data, may be retransmitted via cable or rf to the originator for applications such as catalog ordering, bank-by-cable, educational test scoring, etc.

In typical video applications, the transfer of data from the array into the shift register occurs during the blanked (no data) retrace period. For a high resolution non-interlaced 1024 lines by 1024 pixels per line, simple black and white CRT screen, as depicted in FIG. **9**, this will require sixteen of the 64K devices in parallel, as depicted in FIG. **10**. For most applications the devices' shift registers **20** will feed their output in parallel into a main video shift register **127** which clocks the data out at the dot or pixel clock speed, i.e. cursor sweep speed.

For the CRT of FIG. **9** the noninterlaced pixel sweep speed or dot clock rate is 12 nsec. However, the video memories of FIG. **10** transfer their outputs to the main shift register **127** once every 16 dot clock cycles and the video memories therefore shift their data every 192 nsec, which is much slower than shift register speed of the device, which is about 40 nsec. Sixteen devices are selected to be used so that a standard 16 line address and memory bus, as well as 16 bit microprocessor may be employed. However, this arrangement simplifies the controller circuitry and reduces the display memory to only sixteen parts while ensuring compatability with a 16-bit processor. Further, by employing $\overline{TR}$ to control the release of random output data the data **6** and address **15** busses may be multiplexed to reduce circuitry and costs. Although for data storage requirement, only 4 of these 64k devices in a by 4 mode could be used, or larger volume devices in various modes could also be used.

For a 512 lines by 512 pixels per line, four plane color system, the CRT is depicted in FIG. **11**, and the arrangement of the memory devices in FIG. **12** and the correspondence of planes to pixels in FIG. **13**. FIGS. **11**, **12** and **13** depict a bit-mapped raster scan color graphics medium resolution system for a 512 lines by 512 pixels per line which requires 4 bits per pixel to obtain 16 color choices. The sixteen video memory devices **5** are organized as four banks or planes of four cascaded chips, with each bank or plane linked to an external 4-bit shift register **127**.

On every shift out operation, the registers **127** supply one bit of the 4-bit combination necessary to describe the color of the corresonding pixel. These 4-bit words are then sent to a color look up table or other video circuitry. To do the proper pixel mappings the host processor's data bus, which sends information to the storage devices, must be reorganized to make sure that the 4-bits, coming out of the external 4-bit shift registers, correspond to the same pixel. Also note that the cycle times of the devices on-chip shift registers increases by a factor of four, since the 4-bit external shift registers must be reloaded every 4 pixel clock intervals, rather than the **16** pixel clock intervals of FIGS. **9** and **10**. However, this speed is still within the speed limitations of the on-chip shift register.

The concepts of the invention are useful in communications systems other than video. For example, multiplexed voice (telephone) or digital data is transmitted serially at very high bit rates via microwave or fiber optic transmission channels. This data is similar in format to the serial video data in line **2** or line **106** in FIG. **7**. Accordingly, the memory device **5** as described above is very useful in processing this type of data. The data is written into the memory **5** from the communications link by the serial, sequentially-addressed (auto increment) port, and/or read from the memory **5** to the communications link by this port. That is, the memory **5** and microcomputer **8** can be part of a receiver, a transmitter, a relay station, or a transceiver. Once in the array **10** of the memory **5**, the data is accessed in parallel or random fashion by the microcomputer **8** for utilization by D-to-A or A-to-D converters for telephone systems, by error detection and correction algorithms, demultiplexing or multiplexing various channels, station-select, encrypting or decoding, conversation to formats for local area networks, and the like. Again, employing a device with one control pin having two control functions reduces the complexity and costs for such systems.

Another use of the concepts of the invention is in a microcomputer system employing a magnetic disc for bulk storage. For example, the so-called Winchester disc provides several megabytes of storage which is accessed serially at bit rates of many megabits/second, similar to the video data rates of FIG. **7**. Programs can be down-loaded from disc to memory **5** in large blocks of 64k-bytes or 128k-bytes, then the microcomputer executes from the memory until a give task is completed or interrupted. The contents of memory **5** can be read out and sent to the disc storage via line **2** while another block is being written into memory **5** via input **22**. The concepts of the invention also allow for reduced complexity and costs in these systems.

While this invention has been described with reference to illustrative embodiments, this description is not

intended to be construed in a limiting sense. Various modifications of the illustrative embodiments, as well as other embodiments of the invention, will be apparent to persons skilled in the art upon reference to this description. It is therefore contemplated that the appended claims will cover any such modifications or embodiments as fall within the ture scope of the invention.

What is claimed is:

1. Apparatus for reading data stored in a random access memory array device, said memory array comprising a plurality of memory cells arranged in rows and columns, said device having a single-bit output and a serial multiple-bit output, comprising:

processor means for generating a data output control signal and a row address signal, and for applying said signals to said device, and

means in said device, responsive to said data output control signal and to said row address signal, for applying said stored data to said serial multiple-bit output responsive to said processor means applying said data output control signal to said device prior to applying said row address signal to said device, and for applying said stored data to said single-bit output of said device responsive to said processor means applying said data output control signal a predetermined time after applying said row address signal to said device.

2. The apparatus described in claim 1, wherein said processor means further generates a read signal, and applies said read signal to said memory array device;

and wherein said applying means in said device is further responsive to said read signal so that said stored data is applied to said single-bit output or to said serial multiple-bit output of said device only if said read signal is also applied to said device.

3. A computer system comprising:

(a) video display means having video signal input means for receiving bit-serial video data for display of data on a screen, said video singal input means connected to a video data path,

(b) bit-mapped video memory means including at least one video memory device, each said video memory device constructed in a single semiconductor integrated circuit and having both bit-serial and single-bit access ports, each said memory device comprising:

(i) an array of rows and columns of read/write memory cells,

(ii) addressing means for selecting rows, or selecting rows and columns, in response to address bits applied to address terminals of the device and in response to an address strobe applied to a control terminal of said device,

(iii) serial register means, and means for loading data from a row of said cells selected by said addressing means to said serial register means,

(iv) a serial data output terminal connected to said serial register means, said output terminal connected to said video data path and thereby coupled to said video signal input means,

(v) at least one single-bit data input terminal, and at least one single-bit data output terminal, and means connecting said at least one single-bit data input and data output terminals to a column or columns selected by said addressing means,

(vi) and control means receiving said address strobe and receiving a control signal from another terminal of said device, and selecting serial access responsive to receiving said control signal prior to said address strobe, and selecting single-bit access responsive to receiving said control signal a predetermined time after receiving said address strobe,

(c) and microprocessor means having bit-parallel bus means separate from said video data path, said bus means coupled to said bit-mapped video memory means, said microprocessor means applying addresses to said address terminals of said at least one video memory device via said bus means for selecting said rows for loading to said serial register means and for selecting rows and columns for said single-bit access, said microprocessor means applying data to said at least one data input terminal of the device via said bus means for writing into said memory cells, said data being supplied to said at least one single-bit data terminal during the time that at least some of the bit-serial video data is being applied to said video signal input means via said video data path, and said microprocessor means applying said address strobe and said control signal to said memory means.

4. A system according to claim 3 wherein said video memory means includes a plurality of said video memory devices, and all of said plurality of video memory devices receive addresses from said bus means in parallel,

and further comprising means for coupling the serial data output terminals of said plurality of video memory devices to said video data path.

5. A system according to claim 4 wherein said coupling means includes shift-register means receiving video data from said serial data output terminals of said plurality of video memory devices, and at least one serial data output from said shift register means is coupled to said video signal input means.

6. A system according to claim 5 including first means for clocking said shift register means, and second means for clocking said serial register means of said plurality of video memory devices.

7. A system according to claim 6 wherein said first means for clocking said shift register means has a clock rate much faster than said second means for clocking said serial register means.

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